

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☒	☐ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☐	☒ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	All code was developed by the authors. R and python code developed for and used in this analysis are available upon request from hawthornebeyer@gmail.com .
Data analysis	The majority of geoprocessing in this analysis used open source libraries in R and Python (the 'osgeo', 'numpy' and 'multiprocessing' Python packages, and the 'sp', 'raster' and 'sf' libraries in R), with some additional processing and map-making in ESRI ArcGIS. References ESRI. ArcGIS [GIS software]. Version 10.8. (2022). GDAL/OGR contributors. GDAL/OGR Geospatial Data Abstraction software Library. Open Source Geospatial Foundation (2023) doi:10.5281/zenodo.5884351. R Core Team. A language and environment for statistical computing. R Foundation for Statistical Computing. Vienna, Austria (2013) doi:ISBN3-900051-07-0.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

All input datasets are available from the references cited. The raster datasets arising from this work are available link below and include: (i) the approximately 1km resolution overview raster representing the proportion of the area of each pixel that has the potential for natural regeneration ('prop pnv v1 1km.tif'; 1.1 GB); (ii) the 30m resolution binary predictions (where a value of >0.5 is allocated a value of 1, acknowledging users may prefer to use a different threshold for their context) of areas suitable for natural regeneration (files named with the prefix 'pnv bin 30m'; 2.3 GB); and (iii) the 30m continuous probability predictions of the potential for natural regeneration, stored as integers representing percentages to minimise file sizes (files named with the prefix 'pnv pct 30m'; 9.3 GB). The 30m resolution datasets have been tiled into 10 degree latitude/longitude tiles. <https://zenodo.org/record/7428804>

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender	NA
Reporting on race, ethnicity, or other socially relevant groupings	NA
Population characteristics	NA
Recruitment	NA
Ethics oversight	NA

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

- ☐ Life sciences ☐ Behavioural & social sciences ☒ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	Based on a sample natural forest regrowth, we use machine learning methods to distinguish areas where natural regeneration did or did not occur across the global tropics as a function of a suite of geospatial biophysical and socio-economic variables. To predict potential tropical forest generation across Earth, we select predictor variables spanning local (site) and landscape scales (from 30 m to country level) that are known to influence the potential for tropical forest regrowth. Spatially explicit values of predictor variables were then used to model the relative potential for natural regeneration in the present (2015), and in the near future (2030), assuming that overall conditions from 2000-2016 apply to future scenarios.
Research sample	The analysis was based on a sample of 5.4 M ha of natural regrowth detected by Fagan and colleagues 2022 (https://doi.org/10.1038/s41893-022-00904-w). Based on this sample use machine learning methods to distinguish areas where natural regeneration did or did not occur across the global tropics as a function of a suite of geospatial biophysical and socio-economic variables.
Sampling strategy	The statistical modelling and validation is based on a set of 6 million random points generated within the regeneration and non-regeneration spatial domains defined above, stratified so that each level of the dependent variable was equally represented but with no other form of spatial stratification. These random points were then intersected with the geospatial covariate datasets (see Predictor Variables section in Methodology in manuscript). Points falling outside of the target biomes (tropical and subtropical forest and savannah biomes), falling within excluded land cover categories (water, bare areas, urban areas, sparse vegetation; ESA CCI Landcover for year 2000), or falling within the regeneration or forestry polygons delineated in the Fagan and colleagues 2022 dataset were excluded.

Data collection	NA - predictor variables were selected for their known influence on the potential for tropical forest regrowth. See Predictor variables section of Methodology in main manuscript.
Timing and spatial scale	Data collection was not relevant to our study. The datasets used in this analysis span from 30 m to the country level spatial resolution.
Data exclusions	Any random points having NoData values for any of the covariates were removed, resulting in a final set of 5.8 million records. NoData values can arise due to mismatches in the coverage of datasets, such as along coastlines.
Reproducibility	NA
Randomization	The points within regeneration spatial domains were sampled as spatially random points generated within the forest regeneration polygons using the 'sf' library in R (function 'st sample'). The points in non-regeneration spatial domains were sampled as spatially random points within latitude limits of +/- 25 degrees using the 'random points GCS' function in R, which accounts for the elliptical shape of Earth.
Blinding	Blinding refers to the concealment of group allocation from one or more individuals involved in a study, and is therefore, not relevant to our study which was desktop-based.

Did the study involve field work? ☐ Yes ☒ No

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

- | n/a | Involved in the study |
|-------------------------------------|--|
| ☒ | ☐ Antibodies |
| ☒ | ☐ Eukaryotic cell lines |
| ☒ | ☐ Palaeontology and archaeology |
| ☒ | ☐ Animals and other organisms |
| ☒ | ☐ Clinical data |
| ☒ | ☐ Dual use research of concern |
| ☒ | ☐ Plants |

Methods

- | n/a | Involved in the study |
|-------------------------------------|---|
| ☒ | ☐ ChIP-seq |
| ☒ | ☐ Flow cytometry |
| ☒ | ☐ MRI-based neuroimaging |